Editor-in-Chief
Discover Mental Health

From:
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Dear Editor-in-Chief,

On behalf of my co-author, Professor Md. Jamal Uddin (corresponding author), I am pleased to resubmit our revised manuscript entitled:

"Digital Connectivity and Common Mental Disorders Among Women in Bangladesh: Evidence of Rural–Urban and Spatial Inequities"

for reconsideration in *Discover Mental Health*, following revision after peer review and editorial recommendations, including transfer from *Social Psychiatry and Psychiatric Epidemiology*.

We sincerely thank the Editor and reviewers for their valuable comments, which have significantly improved the quality and clarity of the manuscript.

In response to the reviewers' feedback, we have undertaken substantial revisions. A detailed point-by-point response is provided in the document **"Response to Reviewers.pdf"**. In addition, we have submitted the following revised files for transparency and reproducibility:

- Clean revised manuscript (final version)
- Revised manuscript with tracked changes (green-highlighted additions and updates)
- Response to reviewers document (point-by-point responses)

For ease of evaluation, all modifications in the revised manuscript are clearly marked using **green-colored highlights**, and we have additionally included **page numbers and line numbers** corresponding to each revision. We believe the revisions have strengthened the methodological rigor, clarity, and policy relevance of the study, particularly in highlighting rural–urban and spatial inequalities in women's mental health in Bangladesh.

We sincerely appreciate your consideration and look forward to your feedback.

Sincerely,
Md Salek Miah
On behalf of the Corresponding Author:
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RESPONSE TO REVIEWER 1

RESPONSE TO REVIEWER 1 – COMMENT 1

Sample Flow and Exposure Definition

Reviewer Comment:

The description of the study sample and exclusions needs to be clearer. It is not clear which numbers come from BDHS 2022 and which are due to exclusions in this study. The flow diagram is unclear, and the large number of exclusions is not well explained. The definition of internet use is also inconsistent.

Author Response (Page 37, Figure 1, Page 6-7, Lines 132-144)

We sincerely thank the reviewer for this valuable and important suggestion. We fully agree that clearer reporting of the BDHS 2022 sampling process and the derivation of the analytic sample is essential for transparency, reproducibility, and alignment with DHS reporting standards.

In response to this comment, we have comprehensively revised the Methods section and Figure 1 to ensure consistency with DHS analytical reporting practices and to clearly distinguish between the original survey sample and the analytic samples used in this study.

- First, we now explicitly describe the BDHS 2022 sampling design, consistent with DHS methodology. The survey initially identified 30,358 eligible ever-married women aged 15–49 years, of whom 30,078 completed the interview, yielding a 99.1% response rate. These figures are now clearly reported in the Methods section (**Page 6-7, Lines 132-144**) following standard DHS reporting conventions.
- Second, we clarify that this study used two separate complete-case analytic samples depending on the availability of exposure information, rather than a single sequential exclusion process. This approach is consistent with prior DHS-based secondary analyses where different indicators have different levels of missingness.
- For ICT ownership and ICT usage analyses, a complete-case analytic sample of 18,987 women was used after excluding observations with missing values in outcome variables (PHQ-9 and GAD-7), exposure variables, and relevant covariates.
- For internet use analysis, a separate restricted complete-case sample of 5,486 women was used due to missing internet use information in the BDHS dataset. This reflects the DHS data structure, where certain variables are not uniformly available for all respondents.
- Third, we have revised **Figure 1** to clearly present a structured flow of the BDHS sampling process and to separately display the two analytic pathways (ICT-related analyses and internet use analysis). This revision ensures that exclusions are transparent, non-overlapping, and clearly traceable from the original survey sample.
- Finally, we have standardized the definition of internet use in the Methods section. Internet use is now defined as a binary variable based on self-reported use within the last 12 months, consistent with DHS-style recoding practices. Respondents reporting any use were coded as users (1), while those reporting no use were coded as non-users (0).

These revisions substantially improve clarity, align the manuscript with DHS reporting conventions, and ensure that the sample derivation process is transparent, reproducible, and methodologically consistent.

Changes made:

- Revised Figure 1 to align with DHS-style sampling flow presentation
- Clearly separated original BDHS sample from analytic samples
- Clarified use of two independent complete-case datasets (ICT vs internet use)
- Standardized internet use definition to DHS-compatible binary coding
- Improved Methods section for transparency and reproducibility
- Removed ambiguity regarding exclusion pathways and missing data handling

RESPONSE TO REVIEWER 1 – COMMENT 2

Reviewer Comment:

The term "insufficient data" is not clearly explained. It is unclear what type of data are missing (outcomes, exposures, or covariates). The exclusion of 13,501 women is not sufficiently justified, and the reasons for this large reduction in sample size should be clearly explained.

Author Response ((Page 37, Figure 1, Page 6-7, Lines 132-144)

We sincerely thank the reviewer for this important and constructive comment.

In response to this comment, we have revised the Methods section and Figure 1 to improve transparency and clarity regarding missing data handling, in line with standard DHS secondary analysis reporting practices.

Specifically, we have removed the vague term "insufficient data" and replaced it with a clear description of variable-level missingness. The exclusion of 13,501 observations was due to missing or incomplete information in key study variables required for analysis, including mental health outcomes (PHQ-9 and GAD-7 scores), digital exposure variables (notably last 12 months internet use, ICT ownership, and ICT usage), and relevant covariates used in multivariable regression models.

We further clarify that internet use was not uniformly available for all respondents in the BDHS 2022 dataset, resulting in substantial missingness for this variable. Therefore, complete-case analysis was applied to ensure consistency and validity in model estimation.

The revised Figure 1 now explicitly presents variable-specific sources of missingness rather than a single aggregated exclusion category, thereby improving transparency in the derivation of the analytic sample.

These revisions ensure clearer reporting of the sample selection process and align the manuscript with DHS-based analytical standards.

Changes made:

- Replaced all instances of "insufficient data" with specific variable-level descriptions
- Explicitly included "last 12 months internet use" as a source of missingness
- Clarified reasons for exclusion of 13,501 observations
- Revised **Figure 1** to show variable-specific missingness instead of generic exclusion
- Improved transparency and reproducibility of sample selection process

RESPONSE TO REVIEWER 1 – COMMENT 3

Clarity and Consistency of Internet Use Definition

Reviewer Comment:

The definition of internet use is not clear. The manuscript uses terms such as "never or rarely" and "high frequency," while Figure 1 suggests a different definition (e.g., use in the last 12 months and at least one week of use). It is unclear whether these refer to the same variable or different definitions. Clarification is needed to improve reproducibility.

Author Response (Page 7-8, Lines 160,161, 173-177)

We sincerely thank the reviewer for this important and constructive comment. In response to this comment, we have revised the Methods section and Figure 1 to ensure consistency and clarity in the definition of internet use.

We clarify that internet use was ultimately operationalized as a binary variable based on self-reported timing of use. Respondents who reported never using the internet or using it only rarely prior to the last 12 months were classified as non-users (0), while respondents who reported using the internet within the last 12 months were classified as users (1).

The previously used terms such as "never or rarely" and "high frequency" were removed from the manuscript to avoid ambiguity, as they do not represent separate analytical categories in the final model. Instead, the final analysis relies solely on the dichotomous classification based on last 12 months internet use.

We have also updated Figure 1 to ensure that the flow diagram reflects only this final operational definition, thereby eliminating inconsistencies between the Methods section and the figure.

These revisions improve clarity, ensure internal consistency, and enhance the reproducibility of the study.

Changes made:

- Standardized internet use definition to last 12 months reference period

- Removed inconsistent terms (“never or rarely”, “high frequency”)
- Clarified binary coding: non-user (0) vs user (1)
- Updated Figure 1 to match final definition
- Improved methodological clarity and reproducibility

RESPONSE TO REVIEWER 1 – COMMENT 2

Interpretation of Rural–Urban Differences

Reviewer Comment:

The comparison between rural and urban mental health burden needs to be clarified. The manuscript incorrectly interprets that anxiety and depression are higher in rural populations based on the proportion of cases (66–67% rural). However, this reflects case distribution rather than prevalence. Proper prevalence comparison should use rural and urban denominators separately.

Author Response (Page 13, Lines 276-279)

We sincerely thank the reviewer for this important and insightful comment. We fully agree that the previous interpretation incorrectly described rural–urban differences in terms of prevalence, whereas the reported figures actually represented the distribution of cases by place of residence.

In response to this comment, we have carefully revised the Results section to ensure correct epidemiological interpretation. Specifically, we have removed statements implying higher prevalence of anxiety and depression in rural populations based on case proportions.

The revised manuscript now clearly states that among women who screened positive for anxiety, 66.9% were from rural areas and 33.1% were from urban areas. Similarly, among women who screened positive for depression, 67.4% were from rural areas and 32.6% were from urban areas. We explicitly clarify that these figures represent the distribution of cases by residence and do not reflect prevalence differences between rural and urban populations.

Furthermore, we have revised the interpretation throughout the Results and Discussion sections to avoid ecological misinterpretation. Where appropriate, we now use neutral wording such as “distribution of cases” rather than “higher burden” or “higher prevalence” in rural populations unless supported by regression-based prevalence comparisons.

These revisions ensure correct statistical interpretation and improve the methodological rigor and clarity of the manuscript.

Changes made:

- Removed incorrect interpretation of “higher rural prevalence”
- Corrected Results text to reflect case distribution only

- Added clarification that figures do not represent prevalence differences
- Revised language in Results and Discussion to avoid ecological fallacy
- Improved epidemiological accuracy and interpretation consistency

RESPONSE TO REVIEWER 1 – COMMENT 2

Interpretation of Rural–Urban Differences

Reviewer Comment:

Because a larger proportion of women in Bangladesh live in rural areas, it is expected that more cases will come from rural populations even if the actual prevalence is similar. In Table 1, the difference by residence is not statistically significant, suggesting no meaningful prevalence difference. Therefore, statements claiming higher rural burden should be revised.

Author Response (Page 14-15, Lines 310-333)

We sincerely thank the reviewer for this important epidemiological clarification. We fully agree with the reviewer's interpretation that the observed higher number of cases from rural areas is primarily influenced by the underlying population distribution in Bangladesh, where a larger proportion of women reside in rural settings.

In response to this comment, we have carefully revised the manuscript to ensure correct epidemiological interpretation and consistency with the statistical evidence.

Specifically, we acknowledge that case distribution does not represent prevalence and therefore cannot be used to infer differences in disease burden between rural and urban populations. Accordingly, we have removed all statements suggesting a higher burden of anxiety or depression among rural women based solely on proportional case counts.

Furthermore, as correctly highlighted by the reviewer, the results presented in Table 1 show that differences in anxiety and depression by place of residence are not statistically significant. This supports the interpretation that there is no strong evidence of meaningful differences in prevalence between rural and urban women in this study.

We have revised both the Results and Discussion sections to reflect this interpretation. The manuscript now emphasizes that observed rural–urban differences in case counts are largely driven by population structure rather than true differences in prevalence, and that findings should be interpreted cautiously in an epidemiological context.

These revisions improve the methodological rigor of the manuscript and ensure accurate interpretation of descriptive and inferential findings.

Changes made:

- Explicitly acknowledged population structure effect (rural majority in Bangladesh)

- Removed all statements implying higher rural burden based on case distribution
- Clarified distinction between case distribution and prevalence
- Incorporated non-significant results from Table 1 into interpretation
- Revised Results and Discussion sections for epidemiological accuracy
- Improved alignment with correct cross-sectional interpretation standards

RESPONSE TO REVIEWER 1 – COMMENT 3

Consistency Between Descriptive Table and Spatial Interpretation

Reviewer Comment:

In Table 1, Rajshahi shows the lowest anxiety prevalence and Rangpur shows one of the highest depressions prevalences. However, the spatial analysis and discussion sections inconsistently describe Rangpur as a "low-burden" region. This creates inconsistency between the descriptive table and spatial interpretation and may affect the validity of spatial conclusions.

Author Response (Page 17, Lines 371-379; Table 1; Figures 4–5)

We sincerely thank the reviewer for this important and insightful comment regarding consistency between the descriptive results and spatial interpretation. In response to this comment, we have carefully revised the Results and spatial analysis sections to ensure full consistency with the findings presented in Table 1. Specifically, we clarified that spatial patterns are outcome-specific and should not be generalized across all mental health outcomes. We acknowledge that while Rangpur shows relatively lower digital exposure and lower anxiety prevalence, depression prevalence is not uniformly low across all divisions, and regional patterns vary depending on the outcome considered. Similarly, Rajshahi consistently shows relatively lower anxiety prevalence compared to other divisions, as reflected in **Table 1**. Accordingly, we have removed all statements that inaccurately described Rangpur as a uniformly low-burden region. The revised manuscript now emphasizes spatial heterogeneity across divisions and clearly distinguishes between anxiety and depression patterns rather than assuming uniform regional burden. Furthermore, we clarify that the spatial analysis is descriptive in nature and based on aggregated division-level prevalence estimates. Therefore, the observed spatial patterns should be interpreted cautiously and should not be considered as evidence of uniform regional risk across all mental health outcomes. These revisions ensure consistency between Table 1 and the spatial interpretation and improve the epidemiological rigor and internal coherence of the manuscript.

Changes made:

- Corrected inconsistent interpretation of Rangpur and Rajshahi
- Removed statements implying uniform "low-burden" regions
- Clarified outcome-specific spatial heterogeneity (anxiety vs depression differs)

- Emphasized descriptive nature of spatial analysis
- Aligned spatial interpretation with Table 1 results
- Improved methodological consistency and epidemiological accuracy

RESPONSE TO REVIEWER 1 – COMMENT 4

Exposure Construction and Justification of Internet Use, ICT Ownership, and ICT Usage

Reviewer Comment:

The construction of the exposure variables (internet use, ICT ownership, and ICT usage) needs clearer justification and documentation. The reviewer requested clarification regarding the source of the exposure definitions, rationale for categorization, equal weighting of ICT devices, distinction between digital and traditional media, justification of cut-offs, and potential inconsistencies arising from combining household-level and individual-level measures.

Author Response (Pages 7-8, Lines 158-177, pages :3-4, lines no: 68-75)

We sincerely thank the reviewer for this thoughtful and constructive comment. In response, we have substantially revised the Methods section to clarify the derivation, conceptual framework, and categorization of all exposure variables.

First, we now explicitly state that the exposure variables were constructed using standardized BDHS 2022 questionnaire items and were informed by previous DHS-based studies examining information and communication technologies (ICTs) and women's health outcomes. In particular, our approach to ICT ownership was adapted from previous studies that utilized DHS and MICS datasets by operationalizing household ownership of communication technologies such as television, radio, mobile phone, and computer as indicators of ICT access (**Cardoso & Sorenson, 2017; Nandi et al., 2024**). Relevant references have now been added to the manuscript.

Second, regarding ICT ownership, we acknowledge that different technologies may vary in functionality and interactivity. However, the objective of the ownership variable was not to measure the specific influence of individual devices, but rather to capture overall household access to communication technologies. Consistent with previous DHS-based studies, each device was assigned equal weight and combined into a summary measure representing the presence of ICT resources within the household. We have now clarified this rationale in the Methods section and additionally acknowledged this simplification as a limitation.

Third, we agree that ICT usage primarily reflects exposure to traditional mass media rather than modern internet-based digital engagement. To improve conceptual clarity, we have revised the manuscript to distinguish ICT ownership (technology access), internet use (individual digital connectivity), and ICT usage (frequency of media consumption). We now explicitly state that ICT usage represents exposure to information channels available through television, radio, and newspapers, which remain important communication platforms in Bangladesh and are routinely collected in DHS surveys.

Fourth, we have clarified the rationale for the selected cut-offs. Internet use was classified according to BDHS internet-use reporting categories, where respondents reporting internet use within the last 12 months were categorized as users, while those reporting no use or use before the previous 12 months were categorized as non-users. Similarly, frequent ICT usage was defined according to DHS media exposure categories and reflected regular exposure to at least one media source.

Finally, we acknowledge that ICT ownership combines household-level measures (television and radio ownership) with technologies that may more directly reflect individual access (mobile phone and smartphone ownership). Because BDHS does not provide detailed information on individual access, duration, purpose, or intensity of technology use, ICT ownership should be interpreted as a proxy measure of household digital access rather than individual engagement. This limitation has now been explicitly discussed in the revised limitations section.

These revisions strengthen the transparency, validity, and reproducibility of the exposure measures and provide clearer justification for their use within the DHS framework.

Changes made:

- Added citations to previous DHS/MICS-based ICT studies supporting exposure construction.
  Clarified that ICT ownership measures household access to communication technologies rather than device-specific effects.
- Distinguished conceptually between internet use, ICT ownership, and ICT usage. Added justification for the selected categorization and cut-offs.
- Clarified that ICT usage reflects media exposure rather than exclusively digital engagement.
  Expanded the limitations section to acknowledge equal weighting of devices and the combination of household-level and individual-level measures.

RESPONSE TO REVIEWER 1 – COMMENT 5

Confounder Selection

Reviewer Comment:

The approach used to select confounders requires clarification. Confounders should be selected based on their relationship with both exposure and outcome rather than statistical significance alone. Some included variables may represent mediators. A conceptual framework or DAG would improve transparency.

Author Response (Page 9, Lines 212-224, 226-227, 230-231; Figure 2(38))

We sincerely thank the reviewer for this important methodological comment. We agree that confounder selection should not be based solely on statistical significance testing.

In response, we have revised the Methods section and incorporated a Directed Acyclic Graph (DAG) to provide a transparent conceptual framework for covariate selection. We now explicitly

state that confounders were selected using a combination of **conceptual, theoretical, and statistical considerations**, rather than relying exclusively on p-values from bivariate analyses.

Specifically, the final adjusted models included variables identified from previous literature, epidemiological theory, and contextual knowledge of women's health and digital exposure in Bangladesh. Although chi-square tests were used to explore crude associations, statistical significance was not the sole criterion for inclusion. Several variables were retained in the adjusted models because they were considered theoretically relevant confounders, even when their statistical associations were weak or non-significant.

Regarding variables such as women's decision-making autonomy, contraceptive decision-making, pressure to become pregnant, and IPV justification, we acknowledge the reviewer's concern that alternative causal structures may be possible. However, based on the conceptual framework and existing literature, these variables were considered important social and reproductive determinants that may influence both digital exposure and mental health outcomes. Therefore, they were retained in the adjusted models to reduce potential confounding.

We have revised the manuscript to clearly indicate that the final adjustment strategy was informed by conceptual, theoretical, and statistical criteria collectively, supported by the DAG framework, rather than by p-values alone.

Changes Made

- Added a **DAG-based conceptual framework** illustrating the covariate selection process.
- Revised the confounder selection section to state that variables were selected based on **conceptual, theoretical, and statistical criteria**.
- Clarified that chi-square analyses were conducted for exploratory purposes and were **not the primary basis for confounder selection**.
- Expanded the description of the final adjustment strategy in the Methods section.
- Justified the retention of theoretically important covariates regardless of statistical significance.

Rural–Urban Stratified Analysis and Model Specification

Reviewer Comment:

The model specification for rural–urban stratified analyses is not clearly described. It is unclear whether separate models or interaction terms were used, and a model equation should be provided for clarity.

Author Response (Page 11, Lines 239-241; Table 2)

We sincerely thank the reviewer for this important methodological clarification. We agree that the previous version did not clearly describe the structure of the regression models used for the overall and stratified analyses.

In response, we have revised the Methods section to clearly specify that **separate survey-weighted multivariable logistic regression models were fitted for the overall sample, rural subpopulation, and urban subpopulation**, rather than using interaction terms in a pooled model.

To further improve clarity and reproducibility, we now provide a general model specification indicating how the exposure and covariates were incorporated into each analysis. For each exposure variable (internet use, ICT ownership, and ICT usage), three models were estimated: an overall model, a rural stratified model, and an urban stratified model, all adjusted for the same set of covariates.

The general model structure is now explicitly stated in the Methods section to ensure transparency in the analytical approach and to facilitate interpretation of rural–urban differences.

Changes Made

- Clearly stated that **separate models were used for overall, rural, and urban analyses**.
- Removed ambiguity regarding interaction-term based modeling (not used).
- Added **model specification equation** in the Methods section.
- Clarified that each exposure (internet use, ICT ownership, ICT usage) was analyzed using the same modeling framework.
- Improved transparency and reproducibility of statistical analysis.

RESPONSE TO REVIEWER 1 – COMMENT 7

Causal Language and Interpretation

Reviewer Comment:

The manuscript uses overly causal language (e.g., "increases," "causes," "psychological health price of digitalization"). Given the cross-sectional design, reverse causation is possible and the results should be interpreted as associations rather than causal effects.

Author Response

(Page 2-3, Lines 45-50, page 14-15, line no: 310-333, page no: 19, line no: 408-410, 415, 416, page no: 20, line no: 427,428,429,431, 435, 440-441; Abstract; Results; Discussion; Conclusion)

We sincerely thank the reviewer for this important conceptual and interpretational clarification. We fully agree that, given the cross-sectional design of the BDHS data, causal interpretations are not appropriate and that reverse causality may also be plausible.

In response, we have carefully revised the manuscript to ensure consistent and appropriate use of **association-based language throughout the paper**. Specifically, all causal terms such as

"causes," "increases," "contributes to," and "protective effect" have been replaced with neutral terms such as "associated with," "linked to," or "correlated with."

We also revised the conclusion to remove any causal framing such as "psychological health price of extreme digitalization" and instead emphasize that the findings reflect **observed associations between digital exposure and mental health outcomes**, which may be bidirectional and influenced by unmeasured confounding.

These revisions ensure that the interpretation is fully aligned with the cross-sectional study design and epidemiological standards.

Changes Made

- Replaced all causal terms ("causes," "increases," "leads to," "protects") with **"associated with" or equivalent neutral terms**.
- Revised the **Abstract** to remove causal implications.
- Revised the **Discussion section** to emphasize association and acknowledge **reverse causation possibility**.
- Modified the **Conclusion** by removing the phrase "psychological health price of extreme digitalization" and replacing it with association-based interpretation.
- Added explicit statement acknowledging **cross-sectional limitation and bidirectional relationships**.

RESPONSE TO REVIEWER 1 – COMMENT 8

Interpretation and Wording Needs to Be More Cautious

Reviewer Comment:

The manuscript uses strong language such as "strong evidence" and "clear disparities," which may overstate the findings given the cross-sectional design. The results should be framed more cautiously as associations.

Author Response

We sincerely thank the reviewer for this important suggestion. We agree that the language used in the manuscript in some sections was overly strong and may not fully reflect the limitations of a cross-sectional study design.

In response, we have carefully revised the manuscript to adopt a more cautious and balanced interpretation throughout. Specifically, we have removed or replaced terms such as "strong evidence," "clear disparities," and similar expressions with more appropriate wording such as "suggests," "indicates," and "is associated with."

We have also ensured that all findings are consistently framed as **associations rather than causal relationships**, in line with the study design and statistical approach. These revisions improve the

scientific rigor and ensure that the interpretation does not overstate the strength of the observed relationships.

Changes Made

- Replaced overly strong terms (e.g., "strong evidence," "clear disparities") with **cautious association-based language**.
- Revised wording in the **Abstract, Results, Discussion, and Conclusion** sections.
- Ensured consistent framing of all findings as **associations rather than causal effects**.
- Improved alignment between interpretation and cross-sectional study design.

RESPONSE TO REVIEWER 1 – COMMENT 9

Spatial Analysis Interpretation and Methodological Scope

Reviewer Comment:

The spatial analysis is descriptive and does not provide evidence of clustering or causal spatial patterns. Terms such as "clustering," "protective influence," or "digital paradox" may be overstated. Division-level analysis is also too coarse to capture within-area variation.

Author Response (Page 23, Lines 507-511)

We sincerely thank the reviewer for this important methodological clarification. We fully agree that the spatial analysis presented in the manuscript is **descriptive in nature** and does not constitute formal spatial clustering or causal spatial inference.

In response, we have revised the manuscript to ensure that the spatial findings are interpreted strictly as **descriptive geographic variation in prevalence and digital exposure across administrative divisions**. We have removed or softened terminology that could imply spatial clustering, causal spatial relationships, or protective effects, including phrases such as "clustering," "protective influence," and "digital paradox."

We also acknowledge the limitation that the analysis is based on **division-level aggregation**, which may mask important within-division heterogeneity. This limitation has now been explicitly added to the manuscript to improve transparency.

Changes Made

- Reframed spatial results as **descriptive patterns only**, not clustering or causal spatial relationships.
- Removed/modified terms such as "clustering," "protective influence," and "digital paradox."
- Clarified that findings represent **division-level prevalence estimates only**.

- Added limitation noting **lack of within-division spatial resolution**.
- Revised interpretation in Results, Discussion, and Spatial Analysis sections.

RESPONSE TO REVIEWER 1 – COMMENT 10

Interpretation and Policy Implications

Reviewer Comment:

The discussion and policy implications would benefit from a more nuanced interpretation. Important confounding factors may not be fully captured in the BDHS dataset, some policy implications appear stronger than the evidence supports, and the role of digital exposure should be interpreted more balancedly, recognizing both potential benefits and harms.

Author Response (Pages 23, Lines 500-511)

We sincerely thank the reviewer for this thoughtful and constructive comment. We agree that the interpretation of our findings should be more cautious and should acknowledge both the limitations of the BDHS dataset and the complexity of digital exposure in relation to mental health.

In response, we have substantially revised the Discussion, Limitations, and Policy Implications sections.

First, we now explicitly acknowledge the possibility of **residual confounding**. Although the analyses adjusted for a broad range of socio-demographic, reproductive, household, and geographic factors, the BDHS does not collect information on several potentially relevant factors, including social support, relationship quality, prior mental health conditions, digital literacy, online experiences, and healthcare access. Therefore, residual confounding cannot be excluded and the observed associations should be interpreted cautiously.

Second, we revised the policy implications to ensure they remain proportionate to the study findings. The manuscript now emphasizes that the observed relationships represent **associations rather than causal effects**, and that not all digital exposure indicators demonstrated consistent associations across outcomes and population subgroups. Consequently, policy recommendations are framed as areas warranting further investigation and monitoring rather than definitive intervention targets.

Third, we have adopted a more balanced interpretation of digital exposure throughout the manuscript. The revised discussion now recognizes that digital technologies may provide important benefits, including improved access to information, communication, social support, education, and health services. At the same time, certain patterns of digital engagement may be associated with poorer mental health outcomes. Because the BDHS dataset does not capture the purpose, quality, duration, or context of digital use, the present study cannot determine the mechanisms underlying these associations.

Accordingly, we have reframed the discussion and policy implications to emphasize the need for future longitudinal and context-specific research to better understand how different forms of digital engagement relate to women's mental health in Bangladesh.

These revisions improve the balance, interpretability, and public health relevance of the manuscript while ensuring that conclusions remain consistent with the cross-sectional nature of the study.

Changes made:

- Added explicit discussion of residual confounding due to unavailable psychosocial and contextual variables.
- Clarified that the findings represent associations rather than causal effects.
- Revised policy implications to align more closely with the observed evidence.
- Added a balanced discussion of both potential benefits and challenges of digital technologies.
- Clarified that the BDHS does not capture the purpose, quality, intensity, or context of digital engagement.
- Reframed recommendations to emphasize future research, monitoring, and context-specific interventions rather than causal interpretations.

REVIEWER 3

RESPONSE TO REVIEWER 3 – COMMENT 1

Abstract / Conclusion

Reviewer Comment:

The abstract is generally clear, but the conclusions are somewhat stronger than what the data support. The study shows associations, yet the wording suggests broader implications about digital strategies and mental health improvement. The conclusion should be revised to reflect associations only and avoid implying policy impact beyond what the analysis directly shows.

Author Response (Abstract; Page 2-3, Lines 45-50)

We sincerely thank the reviewer for this valuable and constructive comment. We fully agree that the original abstract conclusion used language that may be interpreted as implying stronger conclusions and policy implications than are supported by the cross-sectional design of this study.

In response to this comment, we have revised the abstract conclusion to ensure that the findings are presented strictly in terms of **associations rather than causal relationships**. The revised version removes language suggesting that digital connectivity or related strategies directly improve women's mental health outcomes.

The conclusion now clearly states that digital connectivity indicators were associated with anxiety and depression among Bangladeshi women, with variations observed across rural–urban residence and geographic regions. We have also ensured that the interpretation is fully aligned with the observational and cross-sectional nature of the BDHS data.

These revisions improve the scientific accuracy, neutrality, and methodological consistency of the manuscript and ensure that the abstract appropriately reflects the evidence generated in this study.

Changes made:

• Revised abstract conclusion to remove causal or policy-effect language

• Replaced overgeneralized implications with association-based interpretation

• Ensured consistency with cross-sectional study design

• Improved alignment between results and abstract conclusion

RESPONSE TO REVIEWER 3 – COMMENT 2

Introduction

Reviewer Comment:

The introduction covers relevant background, but it needs clearer structure and tighter writing. Some sentences are difficult to follow and would benefit from editing for clarity. The transition between global context, ICT definitions, and Bangladesh-specific issues is not smooth. Most importantly, the research gap is not clearly stated. It is not fully clear what this study adds beyond existing work using similar datasets. The reviewer suggests clarifying the contribution and improving readability.

Author Response (Introduction; Page 3-4, Lines 65-75, page : 5, line no: 103-113)

We sincerely thank the reviewer for this valuable and constructive feedback. We agree that the introduction required clearer structure, improved flow, and a more explicit articulation of the research gap.

In response to this comment, we have substantially revised the Introduction section to improve clarity, logical flow, and coherence. Specifically, we have simplified complex sentences and improved readability to ensure smoother transitions between the global context of digital connectivity, the conceptualization of ICT-related measures, and the Bangladesh-specific background.

We have also strengthened the transition between sections to ensure a more logical progression from global evidence to national context, and finally to the specific focus of this study. This revision improves the overall structure and readability of the introduction.

Most importantly, we have clearly redefined and explicitly stated the **research gap and study contribution**. The revised introduction now highlights that, although previous studies have examined ICT or digital exposure in relation to health outcomes using DHS data, limited evidence exists in the Bangladeshi context simultaneously examining multiple dimensions of digital connectivity (internet use, ICT ownership, and ICT usage) in relation to women's mental health outcomes, while also incorporating rural–urban stratification and spatial variation analysis. This study therefore contributes by providing a more comprehensive and multidimensional assessment of digital exposure and mental health outcomes using nationally representative BDHS 2022 data.

These revisions improve the clarity, coherence, and scientific contribution of the Introduction section and better position the study within the existing literature.

Changes made:

• Improved clarity and readability of Introduction text

• Strengthened logical flow between global, conceptual, and Bangladesh-specific context
• Clearly articulated research gap

• Highlighted study novelty and contribution using multidimensional digital exposure measures
• Improved justification for rural–urban and spatial analysis components.

RESPONSE TO REVIEWER 3 – COMMENT 3

Definition of Key Variables (ICT exposure construction)

Reviewer Comment:

The construction of ICT-related exposure variables lacks clarity and justification. The overlap between internet use, ICT ownership, and ICT usage, the equal weighting of heterogeneous devices, and the binary categorization raise concerns about conceptual validity and comparability. The rationale and limitations need clearer explanation.

Author Response (Page 7-8, Lines 160-161 and 173-177 Section 2.4; Figure X)
We sincerely thank the reviewer for this insightful and methodologically important comment. We agree that clearer justification is required to improve the conceptual transparency and interpretability of the ICT-related exposure variables.

In response, we have substantially revised the Methods section (Section 2.4) to clarify the conceptual basis and operationalization of the three exposure variables: internet use, ICT ownership, and ICT usage. These measures were developed to capture complementary but distinct dimensions of digital connectivity, consistent with prior DHS-based and public health literature. In particular, we followed and adapted established approaches from previous studies using DHS/MICS data, including Cardoso and Sorenson (2017) and Nandi et al. (2024), which operationalized ICT exposure using household ownership of multiple devices and frequency-based media use indicators. These studies demonstrate that composite ICT measures are commonly used in large-scale surveys where detailed digital behavior data are not available.

First, internet use captures individual-level active engagement with digital platforms. Second, ICT ownership reflects household-level access to communication technologies (television, radio, mobile phone, smartphone), which serve as proxies for availability of digital infrastructure. Third, ICT usage represents frequency of exposure to information channels (e.g., television, radio, newspapers), reflecting routine media engagement. We acknowledge that these domains are related but conceptually distinct, and were therefore analyzed separately rather than merged into a single index.

We also clarify that all variables were dichotomized to ensure comparability across DHS respondents and to maintain consistency with prior DHS-based studies, where standardized

frequency categories are limited. This approach facilitates interpretation in population-level analyses, although we acknowledge that it reduces variability and may not fully capture intensity, duration, or quality of digital engagement.

Regarding ICT ownership, we recognize that assigning equal weight to different devices (radio, television, mobile phone, smartphone) simplifies heterogeneous technologies with different functionalities. However, this summative approach is widely used in DHS-based literature (including Cardoso and Sorenson, 2017) as a pragmatic method to approximate overall household digital access in the absence of detailed usage data. We have now explicitly acknowledged this limitation in the revised manuscript.

Finally, we have added clarification that combining these indicators inevitably introduces simplification, and that our measure does not capture digital literacy, purpose of use, or intra-household access differences, which should be considered in interpreting the findings.

Changes made:

We revised Section 2.4 to:

- Clearly distinguish internet use, ICT ownership, and ICT usage as separate constructs
- Add justification based on prior DHS and MICS literature
- Clarify rationale for binary categorization and composite scoring approach
- Acknowledge equal-weight limitation in ICT ownership index
- Explicitly state measurement limitations (lack of intensity, purpose, and digital literacy data)
- Improve conceptual clarity and reproducibility of exposure definitions

RESPONSE TO REVIEWER 3 – COMMENT 4

Missing Data and Sample Selection

Reviewer Comment:

A large proportion of observations were excluded due to missing or non-applicable data, particularly for internet use. The justification is not sufficiently clear, and the potential for selection bias is not adequately addressed. The characteristics of excluded participants and the implications for bias should be clarified.

Author Response (Page 12, Lines 252; page no: 24 line no: 522-524)

We sincerely thank the reviewer for this important methodological concern. We fully agree that the large proportion of excluded observations, particularly in the internet use analysis, requires clearer explanation and a more transparent discussion of potential selection bias.

In response, we have revised the Methods section and clarified the sample construction process in Figure 1. The exclusion of observations was primarily driven by the availability of complete information on key variables required for each analytic model. Specifically, internet use was available only for a subset of respondents in the BDHS dataset, resulting in a reduced analytic sample (n = 5,486) for this exposure. Similarly, the ICT ownership and ICT usage analyses were conducted on a separate complete-case sample (n = 18,987), after excluding records with missing exposure, outcome, or covariate information.

We now explicitly clarify that the internet use variable is not universally applicable in the original dataset due to skip patterns and non-response in the DHS questionnaire, which is a known feature of DHS survey design. Therefore, the reduced sample reflects variable-specific data availability rather than arbitrary exclusion.

We also acknowledge the reviewer's concern regarding potential differences between included and excluded participants. While formal comparison of excluded cases is limited by missingness in key exposure variables, we now explicitly state in the Limitations section that the possibility of selection bias cannot be fully ruled out. We have clarified that individuals excluded due to missing ICT or internet data may differ systematically in unobserved characteristics, which could influence the observed associations.

Importantly, we also clarify that all multivariable models were adjusted using confounders selected based on a combination of statistical evidence (bivariate association screening) and strong conceptual and theoretical justification grounded in prior literature. This approach was used to ensure that key socio-demographic and contextual determinants of both digital exposure and mental health outcomes were retained in the final models, even when statistical associations were weak or non-significant.

Changes made:
We revised the manuscript to:
• Clearly explain reason for large exclusion (DHS skip patterns + variable-specific availability)
• Distinguish between internet use and ICT analytic samples
• Clarify complete-case analysis approach for each exposure
• Add explicit acknowledgement of potential selection bias
• Strengthen Limitations section regarding missing data impact
• Clarify that confounders were selected using both conceptual/theoretical and statistical criteria, not p-values alone
• Improve transparency of sample selection in Figure 1

RESPONSE TO REVIEWER 3 – COMMENT 5

Confounder Selection

Reviewer Comment:

The confounder selection strategy is not fully appropriate. Relying mainly on statistical significance may introduce bias. Confounders should be selected based on theory and prior evidence, and the rationale should be clarified.

Author Response (Page 10-11, Lines 212-224, 226-227, 230-231 Section 2.6–2.7; Figure 2)
We sincerely thank the reviewer for this important methodological suggestion. We fully agree that confounder selection should not be driven primarily by statistical significance alone, as this may lead to biased adjustment and model misspecification.

In response, we have revised the Methods section to more clearly describe our covariate selection strategy. We now explicitly clarify that confounders were identified using a combined approach incorporating (i) theoretical and conceptual evidence from prior literature, (ii) a Directed Acyclic

Graph (DAG)-informed framework, and (iii) supplementary statistical screening (bivariate associations) used only for exploratory guidance and not as the primary inclusion criterion.

Specifically, variables included in the final adjusted models were selected based on their established or plausible relationships with both digital exposure (internet use, ICT ownership, ICT usage) and mental health outcomes (anxiety and depression), as documented in prior epidemiological and public health studies in similar settings. These included socio-demographic, reproductive, household, and contextual factors relevant to women's health in Bangladesh.

We further clarify that while chi-square tests were conducted during the initial analysis phase, statistical significance was not used as the main criterion for confounder inclusion. Several variables were retained in the final models even when they were not statistically significant, because they were identified as important confounders in the DAG framework and supported by prior theoretical evidence.

We also acknowledge the reviewer's concern regarding potential over-adjustment. Variables such as empowerment, contraceptive decision-making, and IPV justification were carefully considered within the conceptual framework. Based on existing literature and the DAG structure, these variables were treated as contextual covariates relevant to women's socio-behavioral environment and were included to minimize residual confounding, while recognizing that some may partially reflect downstream social determinants.

Changes made:
We revised the manuscript to:
• Clearly state that confounder selection was not primarily based on p-values
• Add explicit reference to DAG-informed conceptual framework
• Emphasize use of prior literature and theory for covariate selection
• Clarify role of chi-square tests as exploratory only
• Justify inclusion of key socio-behavioral covariates
• Improve transparency and reduce risk of selection bias in model specification

RESPONSE TO REVIEWER 3 – COMMENT 6

Results Presentation and Interpretation

Reviewer Comment:

The results section lacks clarity in distinguishing unadjusted and adjusted findings. Some interpretations (e.g., rural–urban differences) are based on proportions rather than population-level comparisons, and effect sizes may be overstated.

Author Response (Page 13, Lines 276-279)

We sincerely thank the reviewer for this careful and insightful comment. We agree that clearer distinction between unadjusted and adjusted findings, as well as more cautious interpretation of effect sizes, is necessary to improve the clarity and rigor of the Results section.

In response, we have revised the Results section to explicitly differentiate between unadjusted (bivariate) and adjusted (multivariable) findings. We now clearly state that the unadjusted analyses describe crude associations based on chi-square tests, whereas the adjusted models estimate

associations after controlling for theoretically and empirically relevant confounders. We have also improved the narrative flow to avoid implying consistency between these two levels of analysis where differences exist.

Regarding rural–urban and other subgroup comparisons, we have revised the wording to ensure that interpretations are based on appropriate epidemiological measures. Specifically, we clarify that some previously reported percentages reflected the distribution of cases rather than population-based prevalence differences, and we have revised the text to avoid misleading interpretations based solely on case proportions.

We also acknowledge the reviewer's concern regarding effect size interpretation. Although several associations were statistically significant, we agree that the magnitude of effects is generally modest. Therefore, we have revised the language throughout the Results and Discussion sections to avoid overstating the strength of associations. Terms implying large or strong effects have been replaced with more neutral wording such as "modest," "small but statistically significant," or "weak-to-moderate association," where appropriate.

Importantly, we emphasize that the observed changes between unadjusted and adjusted models reflect the influence of confounding control. Some associations that appear weak or non-significant in unadjusted analyses become significant after adjustment due to the presence of suppressor effects and the control of key socio-demographic and contextual variables identified through a combined theoretical, conceptual, and data-informed approach.

Changes made:
We revised the manuscript to:
• Clearly separate unadjusted (bivariate) and adjusted (multivariable) results
• Correct interpretation of rural–urban and subgroup comparisons based on case proportions
• Rephrase overstrong interpretations of effect sizes
• Improve consistency in reporting statistical findings
• Clarify that changes between models reflect confounding adjustment effects
• Ensure cautious and epidemiologically appropriate interpretation throughout the Results section

RESPONSE TO REVIEWER 3 – COMMENT 7

Rural–Urban and Spatial Analysis

Reviewer Comment:

The rural–urban and spatial analysis is largely descriptive. It is unclear whether formal interaction tests were conducted, and the spatial interpretations appear too strong given the analytical approach. The authors should clarify the methods and avoid overinterpretation.

Author Response (Page 17, Lines 371-378; page no: 18, line :392-401, Section 2.9; Figures 4–5)
We sincerely thank the reviewer for this valuable methodological and interpretative comment. We agree that clearer clarification of the analytical approach and more cautious interpretation of spatial patterns are necessary.

In response, we have revised the Methods section to clearly state that rural–urban analyses were conducted using stratified multivariable logistic regression models rather than formal interaction (effect modification) tests. Specifically, separate models were fitted for rural and urban subgroups to estimate associations between digital exposure variables and mental health outcomes within each population stratum. We have now clarified that interaction terms between residence and exposures were not included in the present analysis.

We also acknowledge the reviewer's concern regarding the spatial analysis. We clarify that the spatial outputs presented in Figures 4 and 5 are based on descriptive division-level prevalence estimates and visual mapping of aggregated data. Therefore, these results are intended to illustrate geographic variation rather than to infer formal spatial clustering, causality, or underlying spatial processes.

Accordingly, we have revised the Results and Discussion sections to avoid strong or causal interpretations of spatial patterns. Terms implying clustering, causation, or mechanistic explanations have been removed or softened, and the findings are now framed strictly as descriptive geographic heterogeneity in mental health and digital exposure indicators.

Changes made:
We revised the manuscript to:
• Clarify that separate stratified models (not interaction terms) were used for rural and urban analyses
• Explicitly state that no formal interaction tests were conducted
• Reframe spatial results as descriptive rather than inferential
• Remove causal or mechanistic interpretations from spatial discussion
• Strengthen methodological transparency in Section 2.9
• Ensure consistency between analytical method and interpretation

RESPONSE TO REVIEWER 3 – COMMENT 8

Interpretation of Findings (Causal Language)

Reviewer Comment:

The discussion sometimes implies that digital connectivity leads to poorer mental health, but the cross-sectional design does not support causal inference. Alternative explanations such as reverse causation and unmeasured confounding should be acknowledged.

Author Response (Page 19, Lines 408-410, 415, 416)

We sincerely thank the reviewer for this important and constructive comment. We fully agree that, given the cross-sectional nature of the BDHS data, causal interpretations are not appropriate, and all findings should be framed strictly as associations.

In response, we have carefully revised the Discussion and Conclusions sections to remove causal language throughout the manuscript. Specifically, statements implying that digital connectivity "leads to," "causes," or "drives" poorer mental health outcomes have been replaced with more appropriate terms such as "is associated with" or "is linked to."

We also acknowledge the reviewer's important point regarding alternative explanations. The revised manuscript now explicitly notes that reverse causation is possible, whereby women experiencing anxiety or depression may alter their patterns of internet use or ICT engagement. In addition, we acknowledge that unmeasured confounding factors—such as prior mental health history, social support, relationship quality, online harassment, and digital literacy—may also influence both digital exposure and mental health outcomes, and could not be fully accounted for in the BDHS dataset.

These limitations have been added to the revised Limitations section to ensure a more balanced and methodologically appropriate interpretation of the findings. We have also ensured that all policy and public health implications are now framed cautiously and aligned with the associative nature of the results.

Changes made:
We revised the manuscript to:
• Remove all causal wording (e.g., "causes," "leads to," "drives")
• Replace with associative terminology ("associated with," "linked to")
• Add explicit acknowledgment of reverse causation
• Add discussion of unmeasured confounding factors
• Strengthen the Limitations section on study design constraints
• Align interpretation with cross-sectional analysis framework

RESPONSE TO REVIEWER 3 – COMMENT 9

Limitations

Reviewer Comment:

The limitations section should be expanded to more clearly acknowledge (i) missing data and selection bias, (ii) simplified exposure definitions, (iii) cross-sectional design, and (iv) limited detail on ICT use (type and intensity).

Author Response (Page 24, Lines 530-535;)

We sincerely thank the reviewer for this valuable suggestion. We agree that the limitations section required further expansion to more transparently reflect the methodological constraints of the study.

In response, we have substantially revised and expanded the Limitations section. First, we now explicitly acknowledge the substantial proportion of missing data, particularly for internet use, and clarify that complete-case analysis may introduce potential selection bias. We further note that excluded observations may differ systematically from included participants in unobserved characteristics, which could influence the observed associations.

Second, we have clarified that the exposure variables (internet use, ICT ownership, and ICT usage) are broad, simplified constructs designed for population-level analysis using DHS data. While these measures are consistent with prior literature, we acknowledge that they do not capture important dimensions such as duration, purpose, quality, or context of digital engagement.

Third, we explicitly reinforce that the cross-sectional study design limits causal inference and does not allow temporal relationships between digital exposure and mental health outcomes to be established.

Finally, we highlight that the dataset does not provide detailed information on intensity, frequency beyond broad categories, or qualitative aspects of ICT use, which limits our ability to distinguish between different patterns of digital engagement and their potential differential associations with mental health.

These revisions have been incorporated to ensure a more balanced and transparent presentation of the study's strengths and limitations.

Changes made:
We revised the manuscript to:
• Explicitly acknowledge missing data and potential selection bias
• Clarify limitations of complete-case analysis
• Expand discussion of simplified exposure measures
• Emphasize lack of intensity/type/context of ICT use data
• Strengthen cross-sectional design limitation
• Improve transparency and methodological rigor of the Limitations section

RESPONSE TO REVIEWER 3 – COMMENT 10

Writing and Clarity

Reviewer Comment:

The manuscript requires careful editing for clarity. Some phrases are unclear or awkward, key terms are used inconsistently (e.g., ICT use, ICT usage, ICT exposure, digital connectivity), and the introduction contains repetition that should be streamlined.

Author Response

We sincerely thank the reviewer for this helpful comment. We agree that the clarity, consistency of terminology, and overall readability of the manuscript required improvement.

In response, we have carefully revised the entire manuscript to improve language clarity and flow. Specifically, unclear or awkward expressions (e.g., "very tremendous symptoms of worldwide disability" and "notify equity-focused interventions") have been corrected and replaced with clearer academic wording throughout the text.

We have also standardized key terminology across the manuscript. Terms such as "ICT use," "ICT usage," "ICT exposure," and "digital connectivity" have now been clearly defined in the Methods section, and we consistently use "digital exposure" (with specified subcomponents) throughout the Results and Discussion sections to avoid ambiguity.

In addition, we have streamlined the Introduction by removing repetitive statements related to global mental health burden and ICT expansion, and improved the logical flow between global context, theoretical background, and Bangladesh-specific evidence.

These revisions substantially improve readability, conceptual clarity, and overall coherence of the manuscript.

REVIEWER 4

RESPONSE TO REVIEWER 4 – COMMENT (INTRODUCTION)

Clarity, Consistency, and Regional Focus

Reviewer Comment:

The Introduction needs clearer writing, simplified ICT definitions, removal of redundancy, consistent terminology, improved phrasing, and more focused contextualization (especially Bangladesh rather than broad regional statements).

Author Response (Page 3-4, Lines 69-75; Section 1 Introduction)
We sincerely thank the reviewer for these helpful and detailed comments. We agree that the Introduction required substantial revision to improve clarity, consistency, and focus.

In response, we have carefully revised the Introduction to improve readability and academic tone. First, we corrected unclear and inappropriate wording such as "very tremendous symptoms of worldwide disability" and replaced it with a clear and standard expression ("a major cause of disability worldwide").

Second, we have simplified and streamlined the definitions of ICT-related concepts. The descriptions of ICT ownership and ICT usage have been shortened, clarified, and presented in a more structured manner to avoid repetition and improve readability.

Third, we have ensured consistent use of terminology throughout the manuscript. Terms such as "internet use," "ICT usage," and "digital exposure" are now clearly defined in the Methods section and used consistently throughout the Introduction, Results, and Discussion to avoid ambiguity.

Fourth, we revised unclear phrases such as "where there are minimal resources and little awareness" to more appropriate academic wording.

Finally, we reduced overly broad regional generalizations (e.g., Africa and Asia) and strengthened the focus on Bangladesh by incorporating more context-specific evidence and aligning the background more closely with the study setting.

These revisions improve the clarity, coherence, and scientific focus of the Introduction.

RESPONSE TO REVIEWER 4 – METHODS COMMENTS

Clarity, Sample Definition, Reporting Standards, and Analysis Description

Reviewer Comment:

The Methods section contains issues in sample description, unclear inclusion/exclusion flow, incorrect reporting of STROBE, misclassification of PHQ-9 severity, limited discussion of missing data impact, and vague description of spatial analysis.

Author Response (Page 6-7, Lines 132-144; Section 2 Methods; Figure 1)
We sincerely thank the reviewer for these detailed and constructive comments. We agree that several parts of the Methods section required clarification and correction to improve accuracy, transparency, and reporting quality.

In response, we have carefully revised the Methods section. First, we corrected the grammatical and formatting issues in the description of the analytic sample and clarified that the final sample includes women aged 15–49 years, consistent with the BDHS eligibility criteria. We also corrected the spacing and reporting of sample sizes (e.g., 18,987).

Second, we have improved the description of the sample selection process by clearly explaining the transition from 30,078 interviewed women to the final analytic samples. We now explicitly describe the stepwise exclusions due to missing exposure, outcome, and covariate data, and we have updated Figure 1 to improve transparency of the selection process.

Third, we corrected "STOBE" to "STROBE (Strengthening the Reporting of Observational Studies in Epidemiology)" throughout the manuscript.

Fourth, we revised the terminology used for PHQ-9 scoring. We now correctly describe PHQ-9 ≥10 as indicating moderate to moderately severe depressive symptoms, in line with standard clinical interpretation, and have removed inaccurate wording.

Fifth, we have expanded the discussion of missing data for internet use (n = 13,501) and explicitly acknowledged its potential implications for selection bias in the Limitations section.

Finally, we clarified the spatial analysis description by revising the wording to reflect its descriptive objective. We now clearly state that spatial mapping was used to illustrate geographical variation in mental health outcomes across administrative divisions in Bangladesh, rather than to infer causal or developmental impacts.

These revisions improve methodological clarity, reporting accuracy, and alignment with standard epidemiological practice.

RESPONSE TO REVIEWER 4 – COMMENT (DISCUSSION)

Reviewer Comment:

Please edit language carefully and report in academic English. Phrases like "very tremendous symptoms," "intricated relationships," "higher odd," and "unpleasantly affect" are non-academic. In addition, there is inconsistency between findings and cited literature, and some interpretations (e.g., "digital connectivity leads to vulnerability," "protective influence," and "digital paradox") are too speculative without causal or spatial evidence.

Author Response

We sincerely thank the reviewer for this careful and constructive evaluation of the Discussion section. We agree that several expressions were not sufficiently aligned with standard academic writing and that the interpretation of findings should be more cautious and better grounded in the analytical approach of this study.

In response, we have thoroughly revised the Discussion section to improve academic tone and clarity. Specifically, non-academic or informal expressions such as "very tremendous symptoms," "intricated relationships," "higher odd," and "unpleasantly affect" have been removed and replaced with standard epidemiological and statistical terminology.

We also revised the interpretation of findings to ensure consistency with both the results and existing literature. The Discussion now more clearly distinguishes between our study findings and prior evidence, and provides a more critical explanation for differences across studies, particularly in relation to context, measurement differences, and population characteristics.

Furthermore, we have carefully moderated causal and spatial interpretations throughout the section. Statements implying causality or spatial mechanisms (e.g., "digital connectivity leads to vulnerability" and "protective influence due to low exposure") have been rephrased to reflect association-based interpretations only. Given the cross-sectional and descriptive spatial design, we now explicitly avoid causal or mechanistic claims.

Regarding the term "digital paradox," we have clarified its usage and grounded it more explicitly in existing literature on mixed effects of digital exposure. We also acknowledge that our study does not formally test this concept, and therefore we now present it as an interpretative description rather than a theoretical conclusion.

These revisions improve the scientific rigor, consistency, and readability of the Discussion while ensuring alignment with the study design and analytical limitations.

Changes made:
- Replaced non-academic language with formal epidemiological terminology
- Harmonized interpretation of findings with cited literature
- Removed or revised causal/spatially speculative statements
- Reframed all interpretations as associations rather than causal effects
- Clarified and limited the use of "digital paradox" terminology
- Improved overall clarity and academic tone of the Discussion section

RESPONSE TO REVIEWER – LIMITATIONS / DISCUSSION INTEGRATION

Reviewer Comment:

Although high missing data on internet use is mentioned in the Limitations, its potential impact is not adequately reflected in the Discussion interpretation.

Author Response (Page 20, Lines 440-441)

We sincerely thank the reviewer for this important methodological observation. We agree that simply reporting missing data in the Limitations section is not sufficient, and its implications for interpretation should also be reflected in the Discussion.

In response, we have revised the Discussion to more explicitly acknowledge the potential impact of the substantial missingness in the internet use variable. We now clarify that a considerable proportion of observations were excluded from analyses involving this exposure, which may have reduced statistical power and introduced potential selection bias if the missingness was not completely random.

We further emphasize in the revised manuscript that, due to this limitation, the estimated associations involving internet use should be interpreted with additional caution. The direction and magnitude of these associations may be influenced by differences between included and excluded participants, and therefore should not be interpreted as fully representative of the entire study population.

These additions ensure that the limitations are consistently reflected not only in a dedicated section but also in the interpretation of the study findings.

Changes made:
• Added explicit discussion of missing internet use data in the Discussion section
• Acknowledged potential selection bias due to missingness
• Clarified impact on interpretation of effect estimates
• Strengthened caution in interpreting internet-related associations

RESPONSE TO REVIEWER – RECOMMENDATIONS

Reviewer Comment:

Recommendations are broad and descriptive; consider making them more specific, actionable, and directly linked to the study findings.

Author Response (Page 23, Lines 500-511)

We sincerely thank the reviewer for this valuable suggestion. We agree that the recommendations should be more focused and directly aligned with the empirical findings of the study rather than being broadly descriptive.

In response, we have revised the Recommendations section to ensure that each policy implication is explicitly grounded in the observed associations between digital exposure and mental health outcomes. The revised version avoids general statements and instead emphasizes targeted, evidence-linked actions, particularly in relation to differential patterns observed across digital exposure indicators and socio-demographic subgroups.

We have also reframed the recommendations to reflect the observational nature of the study, ensuring that they are presented as areas for policy consideration and further investigation rather than definitive intervention effects.

These revisions improve the practical relevance, specificity, and scientific alignment of the policy implications with the study findings.

Changes made:
• Replaced broad statements with findings-specific recommendations
• Aligned each recommendation with observed associations in the results
• Avoided overgeneralized policy claims
• Framed recommendations as evidence-informed considerations rather than causal directives